

INTRODUCTION TO SYSTEMS ANALYSIS AND DESIGN: AN AGILE, ITERATIVE APPROACH

SATZINGER | JACKSON | BURD

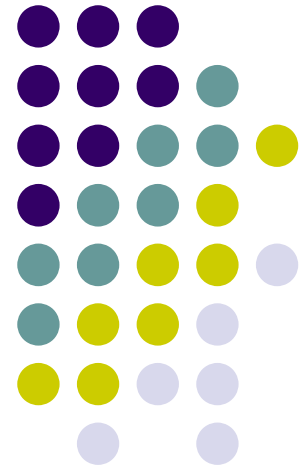
CHAPTER 2

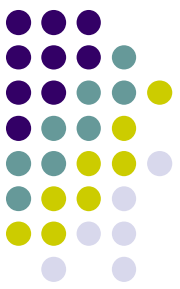
Investigating System Requirements

Chapter 2

Introduction to Systems
Analysis and Design:
An Agile, Interactive Approach
6th Ed

Satzinger, Jackson & Burd

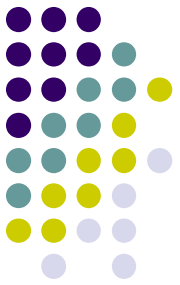




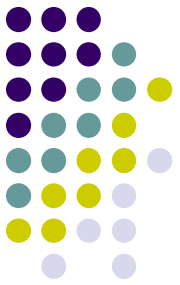
Chapter 2 Outline

- The RMO Consolidated Sales and Marketing System Project （RMO 综合销售项目）
- Systems Analysis Activities （系统分析活动）
- What Are Requirements? （什么是需求）
- Models and Modeling （模型和建模）
- Stakeholders （利益相关者）
- Information-Gathering Techniques （信息收集技术）
- Documenting Workflows with Activity Diagrams （用活动图记录工作流）

Learning Objectives

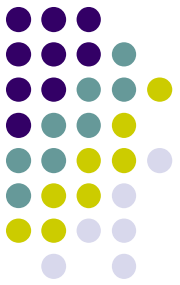


- Describe the activities of systems analysis (描述系统分析的活动)
- Explain the difference between **functional** and **nonfunctional** requirements (解释功能性需求和非功能性需求之间的区别)
- Describe the role of models in systems analysis (描述模型在系统分析中的作用)



Learning Objectives

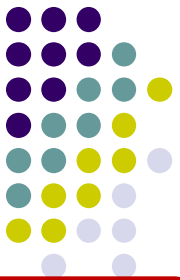
- Identify and understand different kinds of stakeholders and their contributions to requirements definition (利益相关者对需求的不同贡献)
- Describe information-gathering techniques and determine when each is best applied (信息收集技术)
- Develop activity diagrams to model workflows (开发活动图，建模工作流)



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RMO Existing Application Architecture (RMO 现有结构)



- Supply Chain Management (SCM)
 - 5 years old; Java/Oracle
 - Tradeshow system will interface with SCM
- Phone/Mail Order System
 - 12 years old; Visual Studio/SQL
 - Reached capacity;
- Retail Store System
 - Older package solution;
- Customer Support System (CSS)
 - Web based system; evolved over the years, mainly support phone/mail orders

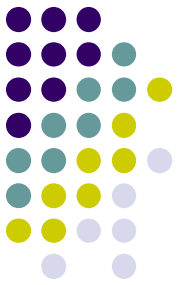
问题：供应链的管理方式发生变化

问题：购物方式发生变化

问题：购物模式发生变化

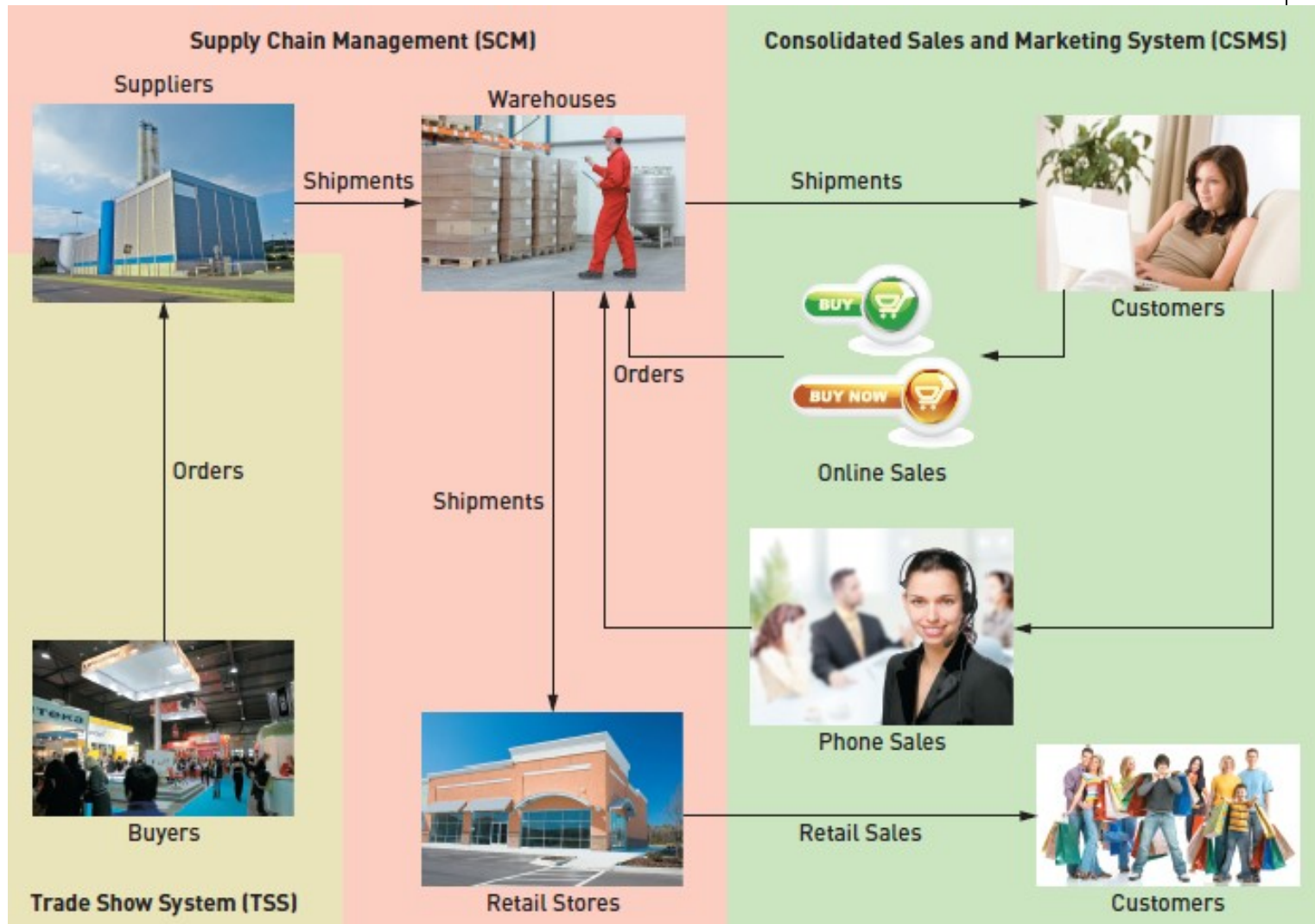
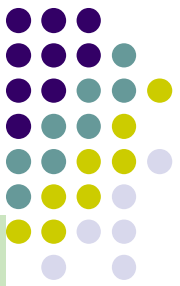
问题：顾客需求发生变化

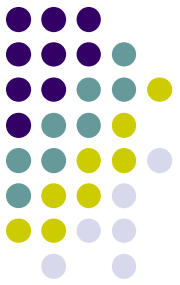
New Consolidated Sales and Marketing System (CSMS , 新的市场推广系统)



- Sales Subsystem (销售子系统)
 - Integrates online, phone, and retail stores
- Order Fulfillment Subsystem (订单实施子系统)
 - Track shipments, rate products and services
- Customer Account Subsystem (顾客账务子系统)
 - Shopping history, linkups, “mountain bucks” rewards
- Marketing Subsystem (市场营销子系统)
 - Promotional packages, partner relationships, more complete merchandise information and reporting

Proposed **Application Architecture:** Integrate SCM and New CSMS



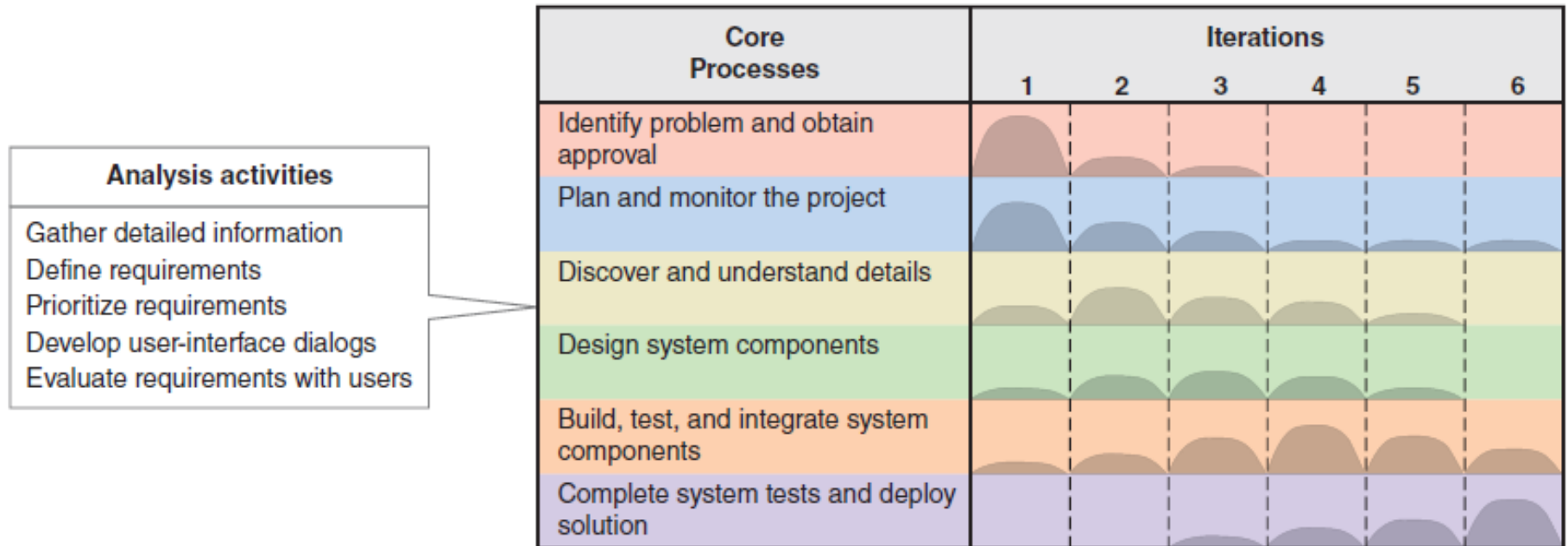
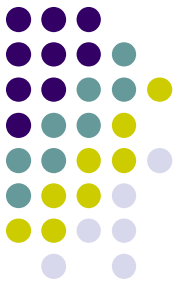


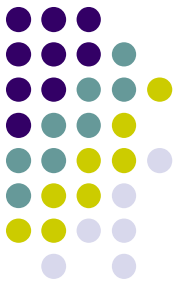
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Systems Analysis Activities

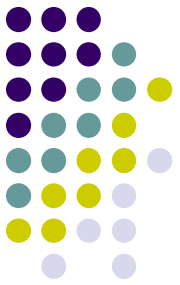
Involve discovery and understanding





Systems Analysis Activities

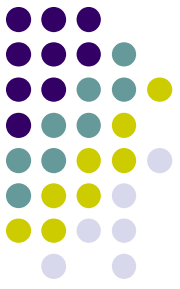
- Gather Detailed Information (收集信息)
 - Interviews, questionnaires, documents, observing business processes, researching vendors, comments and suggestions
- Define Requirements (定义需求)
 - Modeling functional requirements and non-functional requirements
- Prioritize Requirements (优先需求)
 - Essential, important, vs. nice to have
- Develop User-Interface Dialogs (用户界面对话框)
 - Flow of interaction between user and system
- Evaluate Requirements with Users (与用户评估需求)
 - User involvement, feedback, adapt to changes



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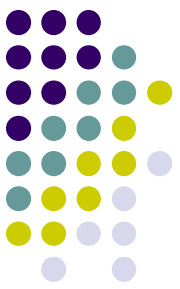
What Are Requirements?



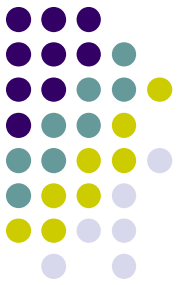
- System Requirements (系统需求) =
 - Functional requirements (功能需求)
 - Non-functional requirements (非功能需求)
- Functional Requirements (功能需求) — the activities the system **must** perform (系统必须执行的活动)
 - Business uses, functions the users carry out
 - Use cases in Chapter 1
- Non-Functional Requirements— other system characteristics (非功能需求——其他系统特征)
 - Constraints and performance goals

FURPS+ Requirements Acronym

识别和分类需求的方法—— FURPS+

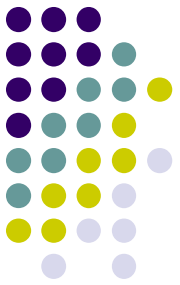


- Functional requirements (功能需求)
- Usability requirements (可用性需求)
- Reliability requirements (可靠性需求)
- Performance requirements (性能需求)
- Security requirements (安全需求)
- + even more categories...



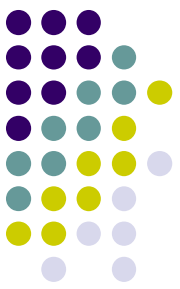
FURPS+ Requirements Acronym

Requirement categories	FURPS + categories	Example requirements
Functional	Functions	Business rules and processes
Nonfunctional 可用性 可靠性 性能 安全 设计限制 开发 接口 物理层 支持	Usability Reliability Performance Security + Design constraints Implementation Interface Physical Support	User interface, ease of use Failure rate, recovery methods Response time, throughput Access controls, encryption Hardware and support software Development tools, protocols Data interchange formats Size, weight, power consumption Installation and updates



Chapter 2 Outline

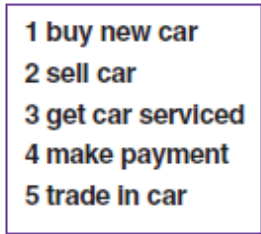
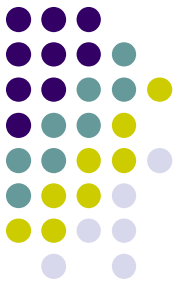
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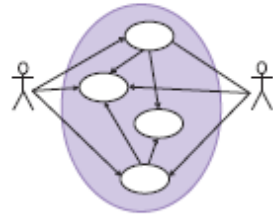
Models and Modeling

- How to **represent** the requirements? 如何表示需求
- Model— a representation of some aspect of the system being built
- Types of Models (模型类型)
 - Textual model (文本) — something written down, described
 - Graphical models (图形) — diagram, schematic
 - Mathematical models (数学公式) — formulas, statistics, algorithms
- Unified Modeling Language (UML , 统一建模语言)
 - Standard graphical modeling symbols/terminology used for information systems

Some Analysis and Design Models



Event list



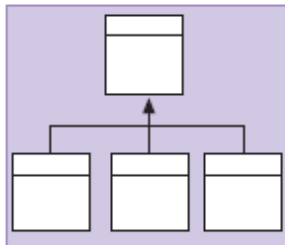
Use case diagram



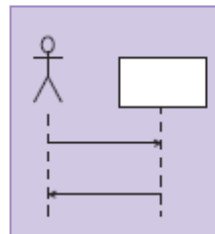
Use case description



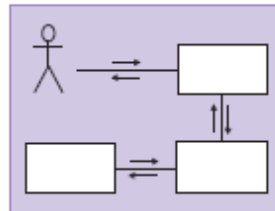
Location diagram



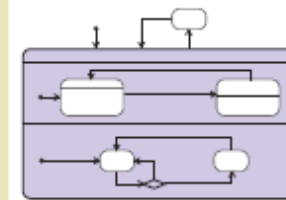
Class diagram



Sequence diagram

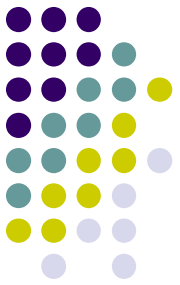


Communication diagram

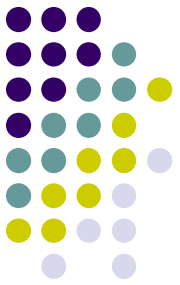


State machine diagram

Reasons for Modeling (建模的原因)



- Learning from the modeling process (从建模过程中学习)
- Reducing complexity by abstraction (通过抽象降低复杂性)
- Remembering all the details (记住所有的细节)
- Communicating with other development team members (与其他成员 / 相关人员沟通)
- Documenting what was done for future maintenance/enhancement (维护、记录)

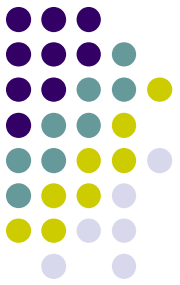


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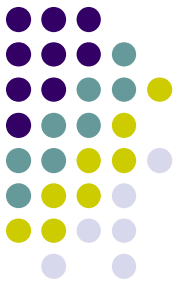
Stakeholders

Who do you involve and talk to?



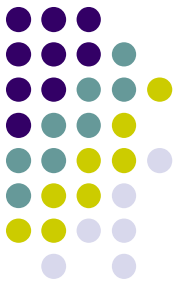
- **Stakeholders** (利益相关者) — persons who have an interest in the successful implementation of the system
- **Internal Stakeholders** (内部利益相关者) — persons within the organization
- **External stakeholders** (外部利益相关者) — persons outside the organization
- **Operational stakeholders** (操作相关者) — persons who regularly interact with the system
- **Executive stakeholders** (执行相关者) — persons who don't directly interact, but use the information or have financial interest

Stakeholders of a comprehensive accounting system for public company



Stakeholders

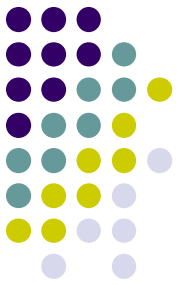
For RMO CSMS Project



- Phone/mail sales order clerks （电话 / 邮件订单员）
- Warehouse and shipping personnel （仓库运输人员）
- Marketing personnel who maintain online catalog information （维护在线目录的营销人员）
- Marketing, sales, and financial managers （市场营销）
- Senior executives （高级管理人员）
- Customers （客户）
- External shippers (e.g., UPS and FedEx) （外部托运人）

Stakeholders

For RMO CSMS Project



Operational

Executive

- Order clerks
- Warehouse and shipping personnel
- Marketing personnel who maintain online catalog information
- Marketing, sales, and financial managers
- Senior executives
- Customers
- External shippers (e.g., UPS and FedEx)

Internal

- Marketing personnel who maintain online catalog information
- Marketing, sales, and financial managers

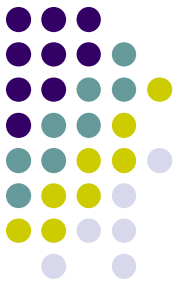
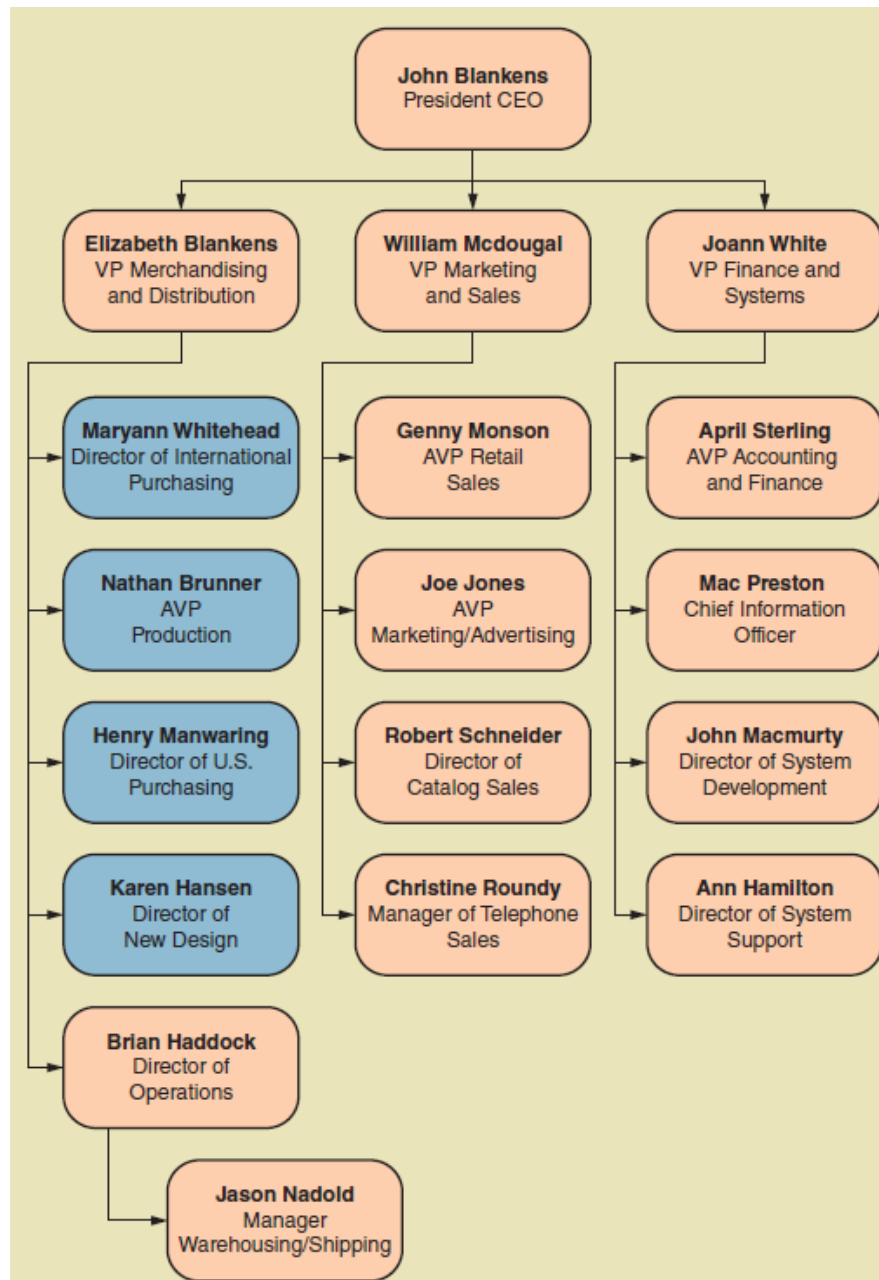
- Senior executives
- Warehouse and shipping personnel

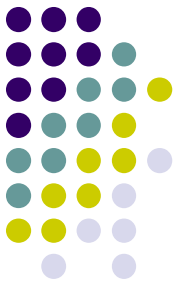
External

- Order clerks
- Customers

- External shippers

RMO Internal Stakeholders

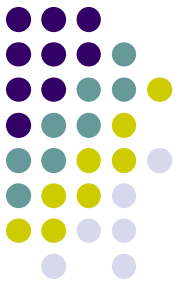




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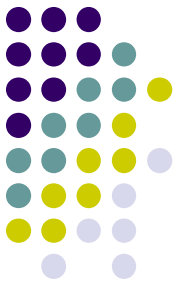
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Information Gathering Techniques



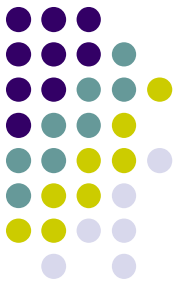
- Interviewing users and other stakeholders （采访用户和其他利益相关者）
- Distributing and collecting questionnaires （发放问卷）
- Reviewing inputs, outputs, and documentation （审查输入、输出和文档）
- Observing and documenting business procedures （观察和记录业务流程）
- Researching vendor solutions （研究供应商解决方案）
- Collecting active user comments and suggestions （收集活跃用户的意见和建议）

Interviewing Users and Other Stakeholders (采访用户)



- Prepare detailed questions (准备详细的问题)
- Obtain and discuss answers to the questions (讨论并获取答案)
- Document the answers (整理答案)
- Follow up as needed in future meetings or interviews (后续跟进)

Themes for Information Gathering Questions



Theme	Questions to users
What are the business operations and processes?	What do you do?
How should those operations be performed?	How do you do it? What steps do you follow? How could they be done differently?
What information is needed to perform those operations?	What information do you use? What inputs do you use? What outputs do you produce?

Distribute and Collect Questionnaires

RMO Questionnaire

This questionnaire is being sent to all telephone-order sales personnel. As you know, RMO is developing a new customer support system for order taking and customer service.

The purpose of this questionnaire is to obtain preliminary information to assist in defining the requirements for the new system. Follow-up discussions will be held to permit everybody to elaborate on the system requirements.

Part I. Answer these questions based on a typical four-hour shift.

1. How many phone calls do you receive? _____
2. How many phone calls are necessary to place an order for a product? _____
3. How many phone calls are for information about RMO products, that is, questions only? _____
4. Estimate how many times during a shift customers request items that are out of stock. _____
5. Of those out-of-stock requests, what percentage of the time does the customer desire to put the item on back order? _____%
6. How many times does a customer try to order from an expired catalog? _____
7. How many times does a customer cancel an order in the middle of the conversation? _____
8. How many times does an order get denied due to bad credit? _____

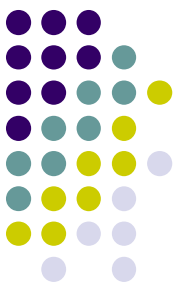
Part II. Circle the appropriate number on the scale from 1 to 7 based on how strongly you agree or disagree with the statement.

Question	Strongly Agree				Strongly Disagree			
It would help me do my job better to have longer descriptions of products available while talking to a customer.	1	2	3	4	5	6	7	
It would help me do my job better if I had the past purchase history of the customer available.	1	2	3	4	5	6	7	
I could provide better service to the customer if I had information about accessories that were appropriate for the items ordered.	1	2	3	4	5	6	7	
The computer response time is slow and causes difficulties in responding to customer requests.	1	2	3	4	5	6	7	

Part III. Please enter your opinions and comments.

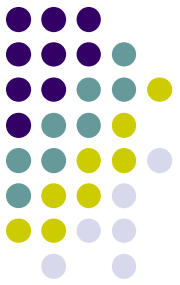
Please briefly identify the problems with the current system that you would like to see resolved in a new system.

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Additional Techniques

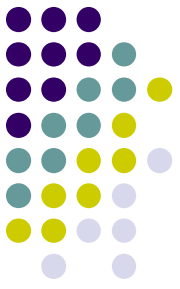
- Observe and Document Business Processes (观察和记录业务流程)
 - Watch and learn
 - Document with Activity diagram (next section)
- Research Vendor Solutions (研究供应商解决方案)
 - See what others have done for similar situations
 - White papers, vendor literature, competitors
- Collect Active User Comments and Suggestions (收集用户反馈)
 - Feedback on models and tests



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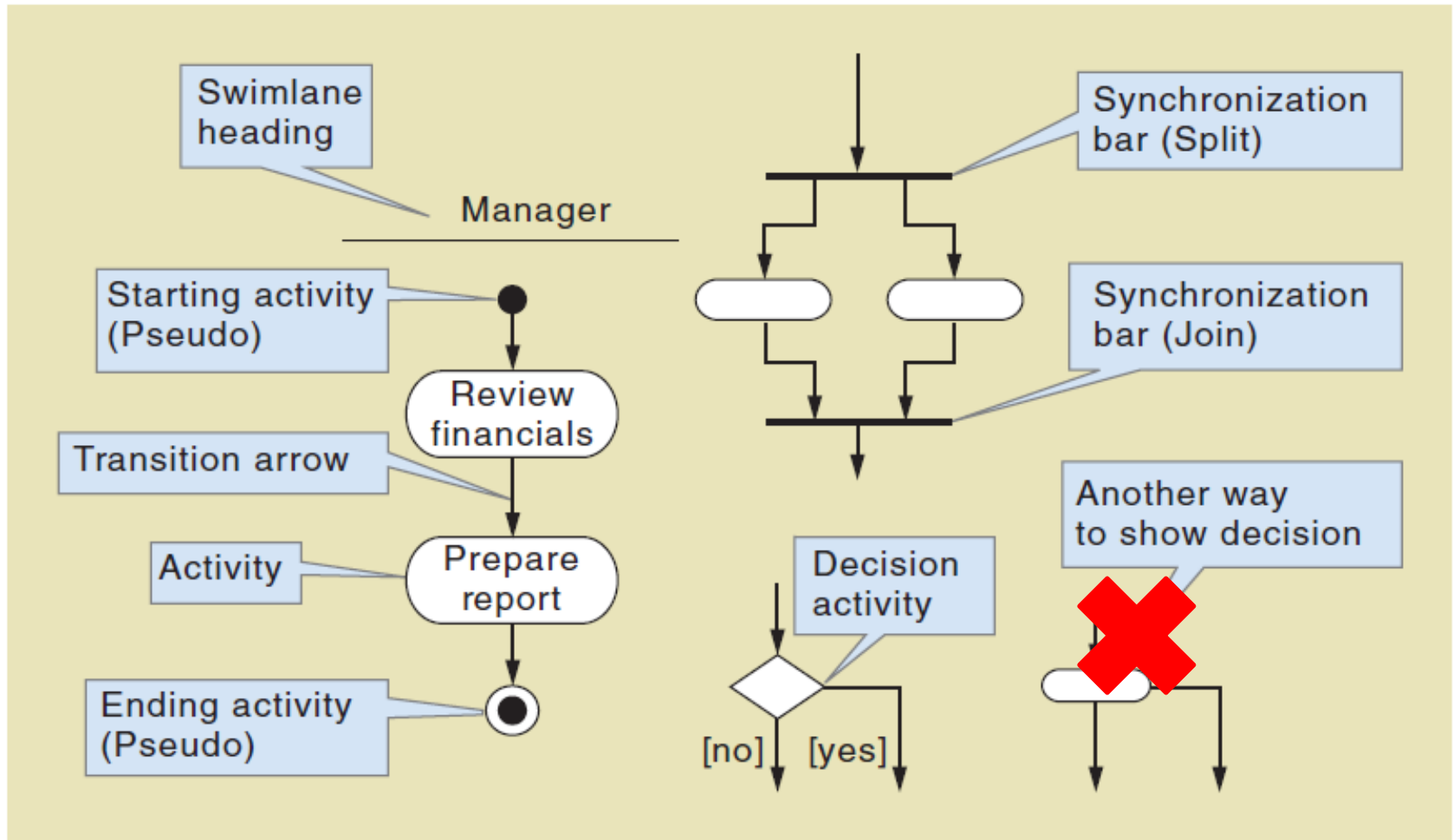
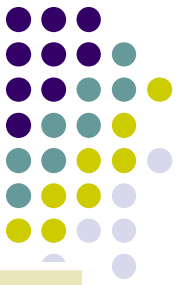
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Documenting Workflows with Activity Diagrams

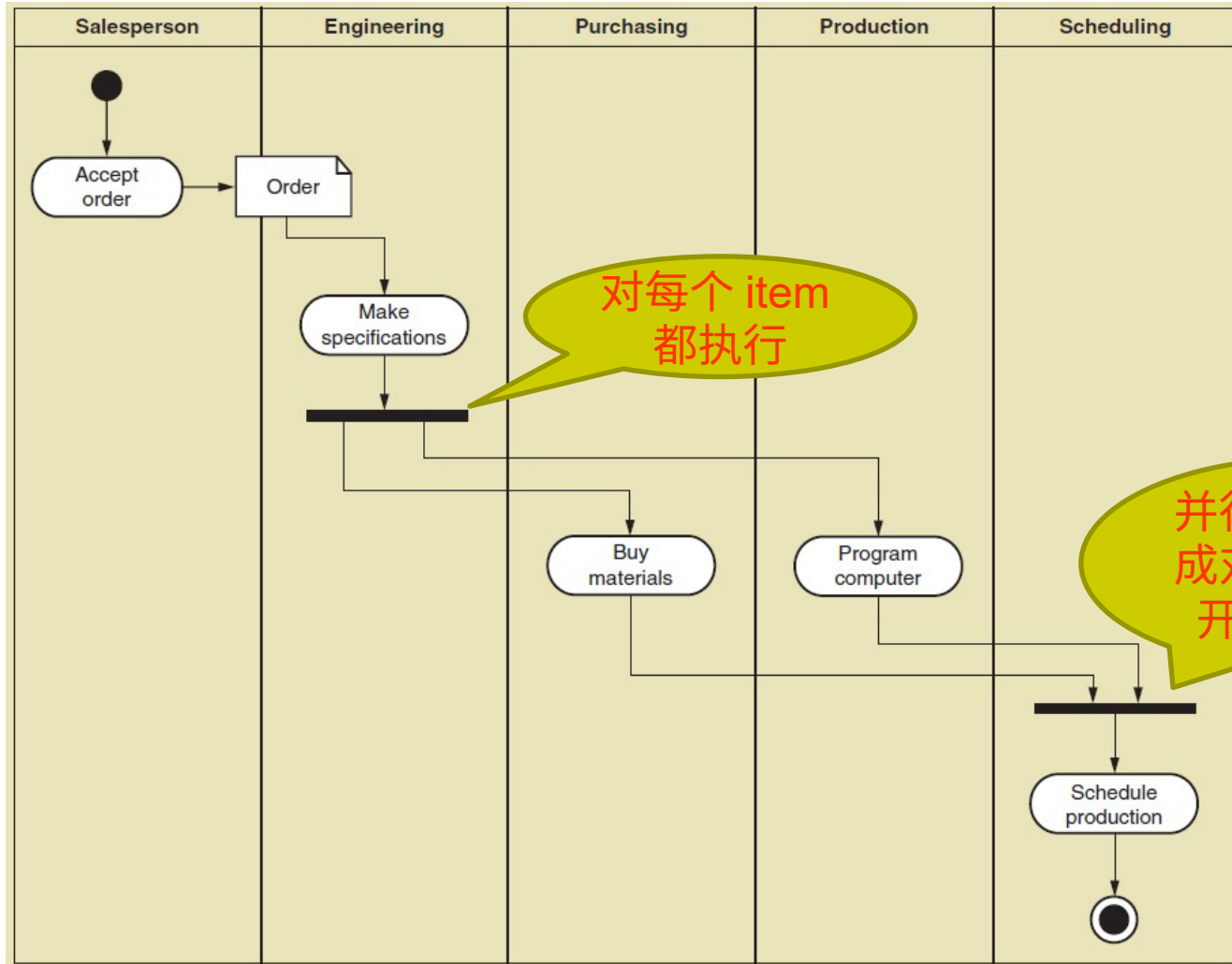
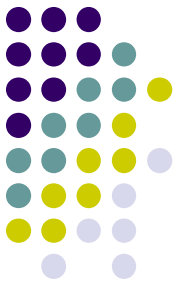


- **Workflow (工作流)** — sequence of processing steps that completely handles one business transaction or customer request (完成处理一个业务的一系列处理步骤)
- **Activity Diagram (活动图)** — describes user (or system) activities, the person who does each activity, and the sequential flow of these activities
 - Useful for showing a graphical model of a workflow (用于显示工作流的图形化模型)
 - A UML diagram

Activity Diagrams Symbols



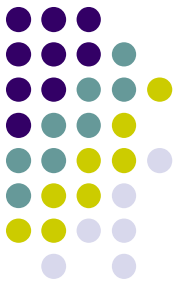
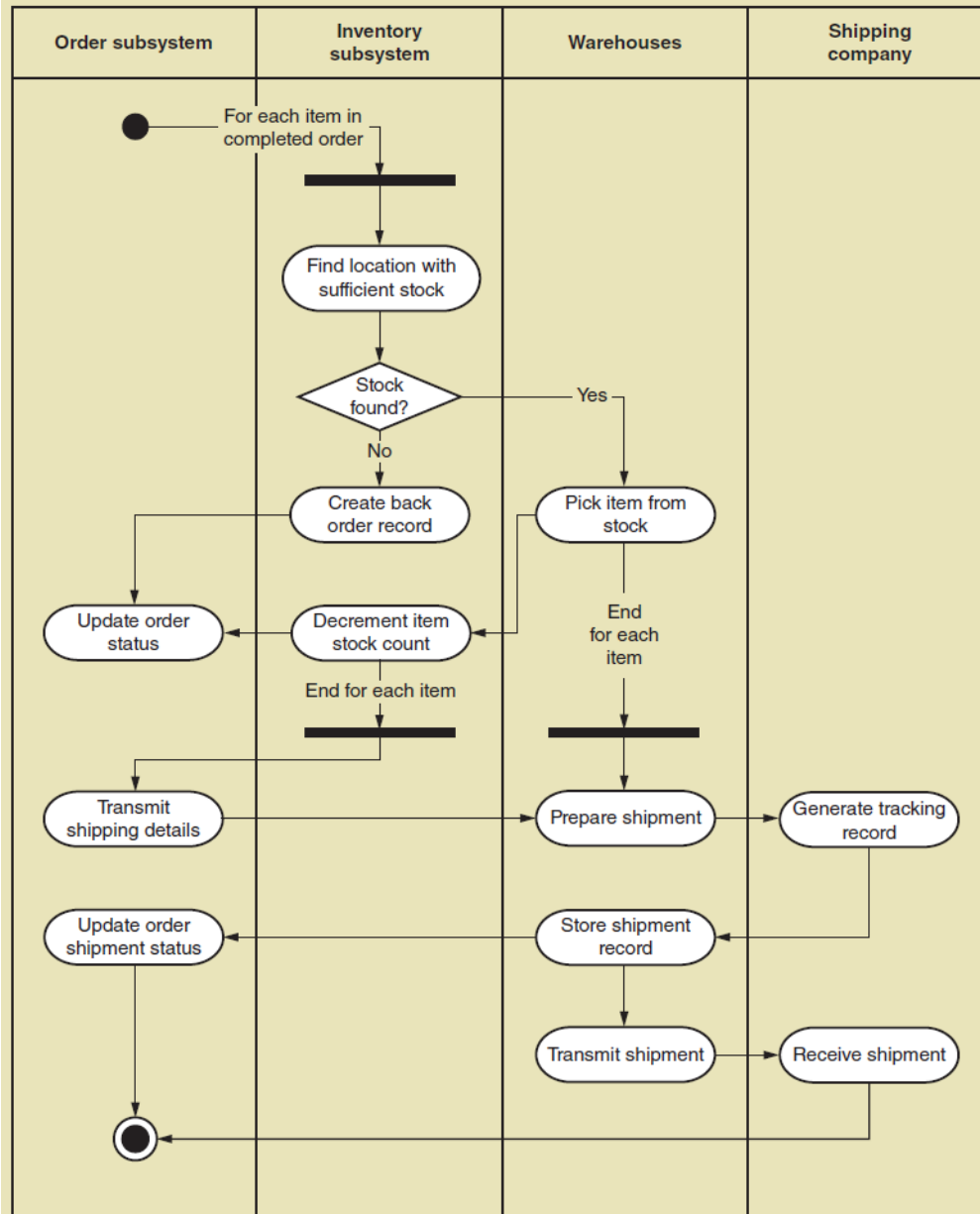
Activity Diagram with Concurrent Paths



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Activity Diagram for RMO Order Fulfillment

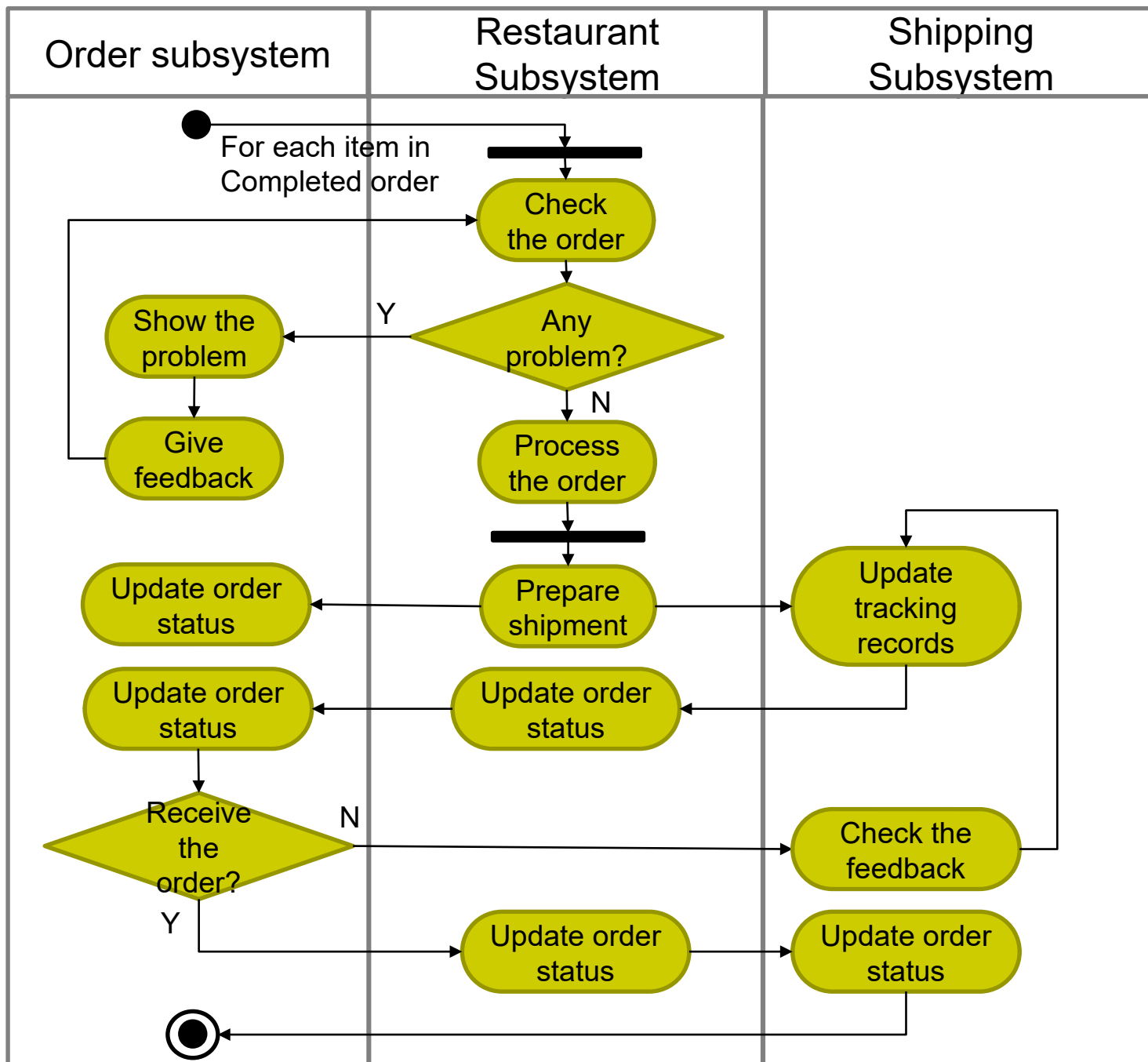
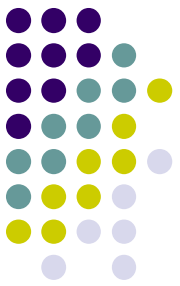
Order Fulfillment

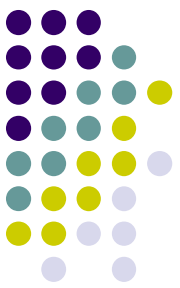




Please analyze the process of ordering a delivery(外卖) from a restaurant website, and then draw an Activity Diagram in UML to reasonably show this process.

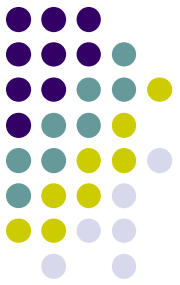
正常使用主观题需 2.0 以上版本雨课堂





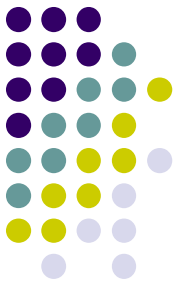
Learning Objectives

- Describe the activities of systems analysis (描述系统分析的活动)
- Explain the difference between functional and nonfunctional requirements (解释功能性需求和非功能性需求之间的区别)
- Describe the role of models in systems analysis (描述相关者对需求定义的贡献)



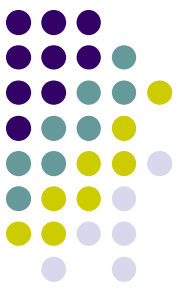
Learning Objectives

- Identify and understand different kinds of stakeholders and their contributions to requirements definition (不同相关者对需求定义均有贡献)
- Describe information-gathering techniques and determine when each is best applied (信息收集技术)
- Develop activity diagrams to model workflows (开发活动图来建模工作流)



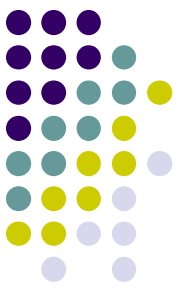
Summary

- Systems analysis involves defining system requirements— **functional** and **non-functional**
- Analysis activities include (分析活动包括)
 - Gather detailed information (收集信息)
 - Define requirements (定义需求)
 - Prioritize requirements (优先需求)
 - Develop user-interface dialogs (用户界面)
 - Evaluate requirements with users (评估需求)
- FURPS+ is the acronym for functional, usability, reliability, performance, and security requirements



Summary

- Models and modeling are used to explore and document requirements （模型和建模用于探索和记录需求）
- A model represents some aspect of a system, and can include textual, graphical, and mathematical models （模型包括文本、图形、数据模型）
- Unified Modeling Language (UML) is the standard set of notations and terminology for information systems models （统一建模语言 UML 是标准集合）



Summary

- Stakeholders are the people who have an interest in the success of the project (利益相关人是对项目成功感兴趣的人)
- There are **internal** vs. **external** stakeholders (内部 vs 外部相关者) and **operational** vs. **executive** stakeholders (运营 vs 执行相关者)
- Information gathering techniques are used to collect information about the project (信息收集)
 - Interviews, questionnaires, reviewing documents, observing business processes, researching vendors, comments and suggestions (访谈、问卷、文件、观察业务流程等)